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ORIGINAL ARTICLE

NFL total quarter back rating (QBR) is not decreased following sport related concussion injury

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Abstract

Hypothesis Our hypothesis was that NFL quarterbacks, due to the high-risk nature of the position and social and financial pressures to return to play following SRC, return to play prematurely and, thus, their on-field performance suffers due to the nature of their injury.

Methods This study was a case–control study of a retrospective cohort of all NFL quarterbacks who were placed on the injury report for a concussion from 2006 to 2019. The data were compiled using publicly available information extracted from NFL transactions. Reportable injuries were based on the criteria that the injury necessitated a team physician referral or emergent care and resulted in scheduled team practice time being missed. Athletes injured in the pre-season or athletes who were injured and did not return to play during the regular were excluded from the study. There were 4 quarterbacks included in the study who suffered repeated concussions during separate seasons; repeated concussion in the same player was treated as two different data points. No quarterback included suffered multiple concussions within the same season. All active NFL quarterbacks injured during a regular season game were eligible for inclusion as age-matched controls. Quarterbacks included in the concussion injury cohort were excluded from the control group. The MEDCALC case–control matching procedure was used to randomly match cases and controls with a 1:1 allocation ratio, respectively, based on age (matching).

Results There was no significant difference in age at injury, years in the NFL when injured, and weeks missed due to injury between the concussion and control groups. There was no statistically significant difference between average season QBR and weekly QBR for the first 6 weeks following return to play after an injury when comparing the concussion to control group.

Conclusion QBR in NFL quarterbacks' did not decrease in the weeks returning from concussion when compared with age-matched control group. This finding was contrary to our hypothesis and suggests that a quarterback's functional skill is not significantly affected by an SRC incident by the time he returns to play.

Level of evidence Level III prognostic study, case–control study.

Keywords NFL · Sport related concussion · QBR · Performance

Introduction

Sport-related concussion (SRC) affects between 1.6 and 3.8 million Americans annually [1, 2].

Among contact sports, football players suffer the highest number of SRCs yearly [3, 4]. The rate of SRC across all collegiate sports rose by 7% per year from 1988 through 2004, and data suggest similar trends in the NFL [3, 5, 6]. Quarterbacks are predisposed to suffering SRC for a number of reasons given the amount of time they possess the ball, potential for blindside hits, high contact loads, and competitive advantage awarded to the defense upon injuring the starting quarterback. During a football game, tackling the quarterback (also known as a sack when behind the line of scrimmage) is a defensive objective throughout the game. Thus, despite a number of recent rule changes, including the targeting rule, aimed to help better protect quarterbacks, NFL quarterbacks remain at a high risk of suffering SRC.

The study was performed at The University of Chicago.

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It has been well established in the literature that concussive injuries can have adverse effects on an athlete's balance and cognitive functioning for the first 24–72 h after injury [7, 8]. These cognitive and balance deficits have been shown to affect a player's on-field performance [8, 9]. Furthermore, establishing recovery from a concussive injury is a difficult task. To date, there is no gold standard for defining a physiological window of time for recovery from a concussion [7, 8]. In addition, recent literature suggests that physiological recovery time from a concussive injury may outlast clinical recovery [7, 8]. High-level athletes, specifically NFL quarterbacks, are often treated with an individualized approach for return-to-play when managing SRC [10, 11]. In these cases, concussive symptomatology as well as modifiers that may prolong recovery, such as a history of prior concussions, migraine, or mood disorders, is considered and the return-to-play protocol is in theory modified accordingly [10]. However, NFL quarterbacks are a unique patient population in that they are subject to high social and financial pressures to return to sport following a concussion injury. Thus, they are high risk for premature return to sport following a concussion and persistent post-concussion symptoms.

There are significant short-term risks associated with premature return-to-sport following a concussion. The majority of SRC injuries resolve within 1–2 weeks, but in some cases, concussion symptoms may persist for months to years beyond the initial injury [12]. Post-concussion syndrome is defined as signs and symptoms of concussion that persist for months after the initial injury [13, 14]. The development of post-concussion syndrome after an initial concussion injury is an area of interest in sports medicine research and our understanding of this topic is currently evolving [12]. Commonly reported post-concussion symptoms include insomnia, headache, dizziness, cognitive intolerance, exercise intolerance, depressed mood, anxiety, irritability, memory loss, poor concentration, fatigue, and noise and light sensitivity [12, 15]. Athletes with persistent concussion symptoms that return to sport are at an increased risk of suffering recurrent or more severe concussions as well as developing prolonged symptom duration [12, 15].

The authors of this paper postulate that NFL quarterbacks, due to the high-risk nature of the position and social and financial pressures to return to play following SRC, return to play prematurely and, thus, their on-field performance suffers due to the nature of their injury. The aim of this study was to investigate the impact of concussion injury on NFL quarterback performance using the well-established total quarterback rating (QBR) as marker for performance. With growing awareness of player safety and the dangers associated with concussions, it is integral to quantify the effect of concussions on player performance in an objective manner. It was hypothesized that NFL quarterback performance would significantly decrease following return to sport after SRC.

Methods

This was a case–control study of a retrospective cohort of all NFL quarterbacks who were placed on the injury report for a concussion from 2006 to 2019. Data were excluded before 2006 because QBR was not calculated prior to that season. The study's aim was to evaluate whether concussion injuries affect quarterback performance following return to play. The data were compiled using publicly available information extracted from NFL transactions, published on a single website (https://www.nfl.com/injuries). Our main source was https://www.nfl.com/injuries, the website publishes a weekly injury report detailing the name of the players injured, their corresponding teams, and a brief description of each injury. This information was cross-referenced with https://www.espn.com/nfl/injuries, https://www.cbssports.com/nfl/injuries/, https://www.pro-football-reference.com/players/injuries.htm, https://bleacherreport.com/nfl-injuries, and each individual team's websites respective injury reports (for example: https://www.chicagobears.com/team/injury-report/, https://www.packers.com/team/injury-report/). To verify each player's date of injury, we cross-referenced each player's reported injury date with an absence of statistics from the respective games they were ruled out for. These methods are in accordance with previously published protocols [16–20].

Reportable injuries were based on the criteria that the injury necessitated a team physician referral or emergent care and resulted in scheduled team practice time being missed. Athletes injured in the pre-season or athletes who were injured and did not return to play during the regular season were excluded from the study. There were 11 quarterbacks who suffered a concussion and did not return to play during the regular season. Of the 11 quarterbacks who sustained a concussion and did not return to play during the regular season, 10 of them were injured within the last 3 weeks of the season. We attributed their absence from returning to play to the timing of their injury. However, 1 quarterback sustained a concussion and missed 11 regular season games. There were 4 quarterbacks included in the study who suffered repeated concussions during separate seasons; repeated concussion in the same player was treated as two different data points. No quarterback included suffered multiple concussions within the same season. Team records, available on each individual team websites, were used to collect demographic information. Our dataset included each player's name, position, team, pathology, date of injury, season, number of weeks missed due to injury,

season QBR, career QBR, and weekly QBR for the first 6 weeks following return to play.

All active NFL quarterbacks injured during a regular season game were eligible for inclusion as age-matched controls. Quarterbacks included in the concussion injury cohort were excluded from the control group. A vigorous online search was performed to determine if any quarterbacks in the control group had suffered a concussion. Our main source of this information was the following websites https://www.nfl.com/injuries, https://www.espn.com/nfl/injuries, https://www.cbssports.com/nfl/injuries/, https://www.pro-football-reference.com/players/injuries.htm, and https://bleacherreport.com/nfl-injuries. The MEDCALC case–control matching procedure was used to randomly match cases and controls with a 1:1 allocation ratio, respectively, based on age (Matching). We choose to age-match controls in an attempt to control for NFL playing experience, and susceptibility to injury related to age.

Statistical analysis was performed using SPSS (version 25.0; SPSS Inc, Chicago IL). The weekly QBR rating data were found to be normally distributed. QBR is a continuous variable, thus, an independent samples *t* test was utilized to assess for a statistically significant difference among the cohorts. All *P* values were two-tailed. *P* values < 0.05 were considered statistically significant. The average weeks missed by NFL quarterbacks and weekly change in QBR for the first 6 weeks following return to play after injury were calculated. The difference between season QBR and weekly QBR following return to play after injury was calculated for both the concussion and control groups. An independent samples *t* test was performed to determine if there was a significant difference between season QBR and weekly QBR following return to play after injury for the concussion versus the control group.

Results

A total of 42 NFL quarterbacks were placed on the injury report for a concussion injury from 2006 to 2019 who met our inclusion criteria. There were 42 NFL quarterbacks included in the age-matched control group. All quarterbacks were injured during a regular season NFL game. Injuries sustained by the age-matched control group included the following: 8 knee injuries, 7 ankle sprains, 5 hamstring strains, 6 shoulder injuries, 4 thumb injuries, 4 foot injuries, 3 elbow injuries, 3 hand injuries, and 2 rib fractures. NFL quarterbacks missed 119 and 98 weeks due to concussion and the above-listed musculoskeletal injuries during the study period. The average age at injury in the concussion versus control group was 27.2 (SD: 3.6) and 27.9 (SD: 4.6), average weeks missed due to injury were 2.83 (SD: 1.9) and 2.33 (SD: 1.6), and average years in the NFL when injured were 5.9 (SD: 3.4) and 6.4 (SD: 4.2) (Table 1). There was no significant difference in age at injury, years in the NFL when injured, and weeks missed due to injury between the concussion and control groups. The average career QBR for the concussion and control groups was 81.2 (SD: 9.1) and 89.1 (SD: 6.8). There was a significant difference in career QBR between the concussion versus the control group.

The average season QBR as well as the average weekly QBR following return to play after an injury is detailed in Table 2. Figure 1 demonstrates the difference between average season and weekly QBR following return to play after injury.

The difference between average season QBR and weekly QBR following return to play after injury is outlined in Table 3. Associated *P* values comparing the difference between average season QBR and weekly QBR following return to play after injury for the concussion versus the control group are listed in Table 3. There was no statistically significant difference between average season QBR and weekly QBR for the first 6 weeks following return to play after an injury when comparing the concussion to control group.

Discussion

The main finding of the current study was that quarterbacks' performances evaluated via QBR did not decrease in weeks returning from concussions when compared to age-matched control. This is contrary to our hypothesis and suggests that a quarterback's functional skill is not significantly affected by an SRC incident by the time he returns to play. This finding is consistent with previous studies evaluating return to play in professional football following concussion injuries [21–23]. Jildeh et al. quantified the effect of concussions on the performance of running backs and wide receivers in professional football players and found players to perform at a similar level in both the acute- and long-term periods after concussion [21]. Reams et al. used defense-adjusted

Table 1 Demographics

Demographics	Concussion injury		Control		
	Mean	Standard deviation	Mean	Standard deviation	*P*-value
Age at injury (year)	27.2	3.6	27.9	4.61	0.4
Years in NFL when injured	5.9	3.4	6.4	4.2	0.51
Weeks missed due to injury	2.8	1.9	2.3	1.6	0.2
Career QBR	81.2	9.1	89.1	6.8	0.01

Table 2 The weekly average QBR following RTP after concussion injury

	Week 1 RTP	Week 2 RTP	Week 3 RTP	Week 4 RTP	Week 5 RTP	Week 6 RTP
Average weekly QBR control	93.4	90.4	88.1	93.3	97.5	88.3
Season average control	91.5	91.5	91.5	91.5	91.5	91.5
Average weekly QBR concussion	84.2	85.9	78.18	77.32	74.02	79.2
Season average concussion	79.96	79.96	79.96	79.96	79.96	79.96

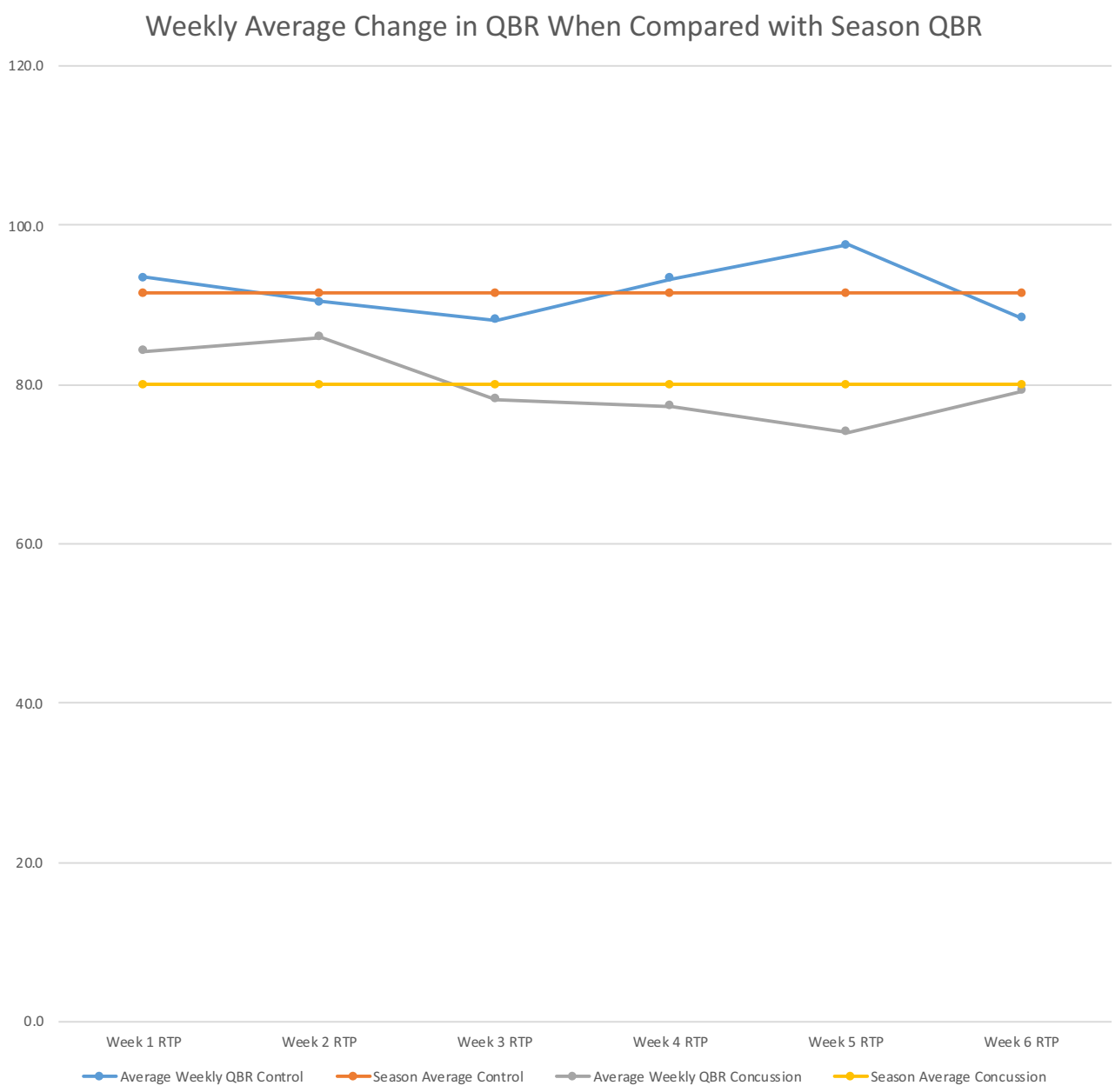

Fig. 1 Difference between season QBR and weekly QBR following RTP after injury

Table 3 The difference in average season and weekly QBR following RTP after injury

	Week 1 RTP	Week 2 RTP	Week 3 RTP	Week 4 RTP	Week 5 RTP	Week 6 RTP
Concussion Injury	4.23	6.02	− 1.78	− 2.64	− 5.94	0.75
Control	1.9	− 1.07	3.37	1.77	5.96	− 3.19
T-Test *P*-Value	0.79	0.11	0.87	0.28	0.18	0.51

yards above replacement as a metric for player performance and found skilled players (running backs, quarterbacks, wide receivers and tight ends) to have no significant difference in performance after returning from concussion injuries when compared to skilled players returning from other head and neck injuries [23]. Kumar et al. similarly found no difference in player performance before and after concussion incidents when using ProFootballFocus performance scores to evaluate NFL player ratings [22]. In contrast, a study performed by Navarro et al. found performance (using fantasy football point scoring) to decrease following concussion incidents [24]. However, in this study, performance was evaluated from the start to the end of the athlete's career and was not specific to the games immediately after return-to-play.

Another finding of the current study was a large number of players in the concussion cohort (20; 47%) were back-up quarterbacks. This finding likely influenced the significant difference noted in career QBR in favor of the age-matched control group. While this was not the focus of the current study and no statistical analyses were done to evaluate the significance of this observation, this may suggest a subset of quarterbacks is at an increased risk for injury. The authors of the current study speculate this may stem from these players' lack of playing experience and ability to avoid dangerous hits or their willingness to perform riskier behaviors to substantiate their talent and earn more playing time. A study by Hannah et al. investigated whether there is a difference in concussion incidence in games between teams who are still "in the playoff hunt" versus games where both teams have had their playoff fate already determined [25]. The hypothesis for higher concussion incidence is predicated on the idea that players make riskier plays and increase the force of their tackles to increase their team's chances of winning. The study found that in general, players are not more likely to suffer concussions in these games. However, at the position-specific level, RBs are more likely to suffer a concussion in an "in the playoff hunt" game [25]. If this trend is further investigated and found to have validity, perhaps, inexperienced players vying for more playing time may require further education and training to avoid devastating hits that cause injury.

Unlike other injuries which can be clearly detected or quantified with modern equipment, technology or tests, the etiology of a concussion occurs at the microscopic level, making it identification particularly difficult with readily available equipment [11, 12, 26, 27]. Immediate detection of SRC is essential to protect athletes from worsening of the injury, increased symptom burden, prolonged recovery, and even increased risk for a subsequent concussion [28–32], which may result in worsening cellular metabolic change and more significant cognitive deficits [12]. Missed concussions could have significant consequence in professional athletes who have economic- and performance-related motivations to remain active [28, 29].

The results of this study do not come without several well-recognized limitations. First, due to the retrospective nature of observational data from a public source, the utilized data were dependent on the reporting of concussions. It is possible that injuries were under-reported for a variety reasons including confidentiality. Second, while the QBR is well-established documented statistic that is often used to assess a quarterback's performance, it may not be a sensitive enough metric to address changes in performance following a concussion. There are a wide variety of quarterbacks that employ a multitude of tactics and styles that may not have equal representation of their performance via QBR. With the new era of advanced metrics, perhaps more sensitive metrics will develop that can better assess a quarterback's cognitive function and decision-making. Additionally, severity of a concussion was not utilized in analysis due to the limited data. If there were trends in performance between performance and concussion severity, the dose-dependent association would not be evident in this study. One quarterback who sustained a concussion during the regular season and missed 11 weeks was excluded from the study due to there being no QBR data from that season to analyze, this was likely a severe injury and this player's exclusion underscores the potential impact of concussion injuries on QBR, our results should be interpreted with this in mind. Another limitation of the study is that while concussion remains a common injury in football, the study population remains relatively small because few quarterbacks play significant minutes per season. This may have led to the inability to obtain significance due to the low number of quarterbacks. A further limitation is due to the inability to incorporate concomitant injuries into the analysis. While, most of these injuries do not prevent participation, they may affect performance which may have confounded the findings of the current study. Finally, it is possible that concussions occurring at different times in the season could adversely affect quarterback performance. However, assessment of performance

with QBR following a concussion was impossible to perform once the season ended.

Conclusion

NFL quarterbacks' performances evaluated via QBR did not decrease in weeks returning from concussions when compared to age-matched controls. This finding was contrary to our hypothesis and suggests that a quarterback's functional skill is not significantly affected by an SRC incident by the time he returns to play. A large number of players in the study were back-up quarterbacks. While this was not the focus of the current study and no statistical analyses were done to evaluate the significance of this observation, this may suggest a subset of quarterbacks is at an increased risk for injury. The authors of the current study speculate this may stem from these players' lack of experience and ability to avoid dangerous hits or their willingness to perform riskier behaviors to substantiate their talent and earn more playing time.

Data availability Raw data were generated at NFL.com. Derived data supporting the findings of this study are available from the corresponding author H.B. on request.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval The article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent For this type of study formal consent is not required.

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